

Statistics for Beginners: Complete Guide

Welcome

Welcome! I understand you don't have a math background, so I'll explain everything from the absolute basics. No prior knowledge needed. Let's start with the very foundations and build up step by step.

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1 What is Statistics?

Statistics is the science of collecting, organizing, analyzing, and interpreting data.

Think of it this way:

- You collect information (data)
- You organize it so it makes sense
- You analyze it to find patterns
- You interpret what those patterns mean

Real-life example:

- Your teacher wants to know how well the class did on a test
- She collects all the scores (data)
- She finds the average score
- She sees how many students passed
- She interprets: “Most students understood the topic”

2 Understanding Data

2.1 What is Data?

Data is just information. It can be:

- Numbers (like test scores, heights, ages)
- Categories (like colors, names, gender)

2.2 Types of Data

Type	Description	Example
Raw Data	Unorganized list of numbers	5, 7, 3, 9, 5, 8
Grouped Data	Organized into groups/intervals	0–10: 5 students, 10–20: 8 students
Ungrouped Data	Individual values listed	5, 7, 9, 11, 13

2.3 Important Symbols You’ll See

Symbol	Name	Meaning
x	Variable	A single data point
n	Sample size	Number of observations
Σ	Sigma	Sum of all values
$\bar{x}$	x-bar	Mean (average)
σ^2	Sigma squared	Population variance
σ	Sigma	Population standard deviation

3 Measures of Central Tendency (Finding the “Center”)

Measures of central tendency tell us the “center” or “typical” value of a dataset. There are three main ones: **Mean**, **Median**, and **Mode**.

3.1 Mean (The Average)

This is what most people call “the average.” You add all numbers and divide by how many numbers you have.

3.1.1 Mean for Raw Data

Formula:

$$\text{Mean} = \frac{\text{Sum of all numbers}}{\text{Total count of numbers}}$$

Example: Find the mean of: 10, 12, 15, 18, 20

Step 1: $10 + 12 + 15 + 18 + 20 = 75$

Step 2: There are 5 numbers

Step 3: $75 \div 5 = 15$

Mean = 15.

Simple explanation: If everyone had the same amount, each person would get 15.

3.1.2 Weighted Mean (Some Numbers Matter More)

When some numbers are more important than others, we use weighted mean.

Example: Your course grade has:

- Homework (30%): Score 80
- Midterm (30%): Score 75
- Final (40%): Score 90

$$\text{Weighted Mean} = (0.30 \times 80) + (0.30 \times 75) + (0.40 \times 90) = 24 + 22.5 + 36 = 82.5$$

Simple explanation: The final exam matters more (40%), so it pulls the average up.

3.1.3 Combined Mean (Average of Two Groups)

When you have two groups and want to find the overall average.

Example:

- Class A: 30 students, average 72
- Class B: 20 students, average 80

$$\text{Combined Mean} = \frac{30 \times 72 + 20 \times 80}{30 + 20} = \frac{2160 + 1600}{50} = \frac{3760}{50} = 75.2$$

Simple explanation: If you combine both classes, the overall average is 75.2.

3.1.4 Mean for Grouped Data (When Data is in Tables)

When data is grouped into intervals, we use the midpoint of each group.

Example:

Class Interval	Frequency (f_i)	Midpoint (x_i)	$f_i \times x_i$
0–10	5	5	25
10–20	8	15	120
20–30	12	25	300
30–40	7	35	245
40–50	3	45	135
Total	35		825

$$\text{Mean} = \frac{825}{35} = 23.57$$

Step-by-step:

1. Find midpoint of each class: (Lower + Upper)/2
2. Multiply midpoint $\times$ frequency
3. Add all these values
4. Divide by total frequency

3.2 Median (The Middle Value)

The median is the middle number when data is sorted from smallest to largest.

3.2.1 Median for Raw Data

- **If n is odd:** Median = value at position $(n + 1)/2$
- **If n is even:** Median = average of two middle values

Example 1 (odd): 2, 5, 7, 9, 12

Sorted: 2, 5, 7, 9, 12

$$n = 5 \text{ (odd)} \quad \text{Position} = (5 + 1)/2 = 3\text{rd number}$$

$$\text{Median} = 7$$

Example 2 (even): 3, 6, 8, 10, 12, 15

Sorted: 3, 6, 8, 10, 12, 15

$$n = 6 \text{ (even)} \quad \text{Positions: } n/2 = 3\text{rd number (8) and } (n/2 + 1) = 4\text{th number (10)}$$

$$\text{Median} = (8 + 10)/2 = 9$$

Simple explanation: The median is the “middle person” when everyone stands in a line.

3.2.2 Median for Grouped Data

Formula:

$$\text{Median} = L + \left[\frac{N/2 - cf}{f} \right] \times h$$

where:

- L = Lower limit of the class containing the median
- N = Total frequency
- cf = Cumulative frequency before the median class
- f = Frequency of the median class
- h = Class width

Example:

Class	Frequency	Cumulative Frequency
0–10	5	5
10–20	8	13
20–30	12	25
30–40	7	32
40–50	3	35

$$N = 35, N/2 = 17.5$$

$$\text{Median class: } 20 - 30 \quad (L = 20, cf = 13, f = 12, h = 10)$$

$$\text{Median} = 20 + \left[\frac{17.5 - 13}{12} \right] \times 10 = 20 + \frac{4.5}{12} \times 10 = 20 + 3.75 = 23.75$$

Simple explanation: We find which group contains the middle value, then estimate exactly where it falls.

3.3 Mode (The Most Frequent Value)

The mode is the value that appears most often.

3.3.1 Mode for Raw Data

Example: 2, 5, 5, 7, 8, 5, 9, 3

$$5 \text{ appears 3 times (most)} \Rightarrow \text{Mode} = 5$$

Special cases:

- **Unimodal:** One mode (like above)
- **Bimodal:** Two modes (2, 3, 3, 5, 5, 7)
- **Multimodal:** More than two modes
- **No mode:** All values appear once

3.3.2 Mode for Grouped Data

Formula:

$$\text{Mode} = L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h$$

where:

- L = Lower limit of modal class (highest frequency)
- f_1 = Frequency of modal class
- f_0 = Frequency of class before modal class
- f_2 = Frequency of class after modal class
- h = Class width

Example:

Class	Frequency
0–10	5
10–20	8
20–30	12
30–40	7
40–50	3

Modal class: 20 – 30 ($L = 20$, $f_1 = 12$, $f_0 = 8$, $f_2 = 7$, $h = 10$)

$$\text{Mode} = 20 + \left[\frac{12 - 8}{2 \times 12 - 8 - 7} \right] \times 10 = 20 + \frac{4}{24 - 15} \times 10 = 20 + \frac{4}{9} \times 10 = 20 + 4.44 = 24.44$$

Important Relationship (Empirical Formula):

$$\text{Mode} \approx 3 \times \text{Median} - 2 \times \text{Mean}$$

This works for moderately skewed distributions.

3.4 Quick Comparison of Mean, Median, Mode

Measure	What it Tells Us	Best Used When
Mean	Average	Data is symmetric, no outliers
Median	Middle value	Data has outliers, skewed data
Mode	Most common value	Data is categorical, want typical value

Example of why all three matter:

Data: 1, 2, 2, 3, 100

Mean = 21.6 (pulled up by 100)

Median = 2 (middle value)

Mode = 2 (most common)

The mean doesn't represent the data well; the median and mode are better here.

4 Measures of Dispersion (How Spread Out Data Is)

Measures of central tendency tell us where the center is. But two datasets can have the same center and be very different!

Example:

Dataset A: 5, 6, 7, 8, 9 (mean = 7)

Dataset B: 1, 4, 7, 10, 13 (mean = 7)

Both have mean 7, but B is much more spread out!

Measures of dispersion tell us **how spread out** the data is.

4.1 Range (Simplest Measure)

Formula:

$$\text{Range} = \text{Maximum value} - \text{Minimum value}$$

Example: 5, 12, 8, 21, 3, 15

$$\text{Maximum} = 21, \text{ Minimum} = 3 \Rightarrow \text{Range} = 21 - 3 = 18$$

Simple explanation: The range tells us the span covered by the data.

Coefficient of Range:

$$\text{Coefficient of Range} = \frac{\text{Max} - \text{Min}}{\text{Max} + \text{Min}}$$

4.2 Quartile Deviation (Using Quartiles)

Quartiles divide data into four equal parts:

- **Q1 (First Quartile):** 25% of data is below this
- **Q2 (Second Quartile):** The median (50% below)
- **Q3 (Third Quartile):** 75% of data is below this

Quartile Deviation (QD):

$$QD = \frac{Q_3 - Q_1}{2}$$

Coefficient of QD:

$$\text{Coefficient of QD} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Example: Find Q1, Q3, and QD for: 3, 6, 7, 8, 9, 11, 13, 15, 18

$$n = 9$$

$$Q_1 \text{ position} = \frac{9 + 1}{4} = 2.5 \Rightarrow Q_1 = \frac{6 + 7}{2} = 6.5$$

$$Q_3 \text{ position} = \frac{3(9 + 1)}{4} = 7.5 \Rightarrow Q_3 = \frac{13 + 15}{2} = 14$$

$$QD = \frac{14 - 6.5}{2} = 3.75$$

Simple explanation: Quartile deviation tells us the average spread of the middle 50% of data.

4.3 Mean Deviation (Average Distance from Center)

Mean deviation is the average distance from the mean (or median).

Formula:

$$MD = \frac{\sum |x_i - \bar{x}|}{n}$$

where $|x_i - \bar{x}|$ means absolute value (ignore negative signs).

Example: 5, 8, 10, 12, 15

$$\text{Mean} = \frac{5 + 8 + 10 + 12 + 15}{5} = 10$$

Deviations from mean:

$$|5 - 10| = 5, |8 - 10| = 2, |10 - 10| = 0, |12 - 10| = 2, |15 - 10| = 5$$

Sum of deviations = $5 + 2 + 0 + 2 + 5 = 14$

$$MD = \frac{14}{5} = 2.8$$

Simple explanation: On average, each value is 2.8 units away from the mean.

Coefficient of MD:

$$\text{Coefficient of MD} = \frac{MD}{\text{Mean}}$$

4.4 Variance and Standard Deviation (Most Important!)

Variance is the average of squared deviations from the mean. **Standard deviation** is the square root of variance.

Why squared deviations?

- Squaring gets rid of negative signs
- Squaring gives more weight to larger deviations

4.4.1 For Raw Data (Ungrouped)

$$\text{Population variance } (\sigma^2) : \sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

$$\text{Population standard deviation } (\sigma) : \sigma = \sqrt{\sigma^2}$$

$$\text{Sample variance } (s^2) : s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

$$\text{Sample standard deviation } (s) : s = \sqrt{s^2}$$

Step-by-step example: Find variance and standard deviation for: 10, 12, 15, 18,

20

$$\text{Step 1: } \bar{x} = \frac{10 + 12 + 15 + 18 + 20}{5} = \frac{75}{5} = 15$$

Step 2: Deviations squared:

$$10 : (10 - 15)^2 = (-5)^2 = 25$$

$$12 : (12 - 15)^2 = (-3)^2 = 9$$

$$15 : (15 - 15)^2 = 0^2 = 0$$

$$18 : (18 - 15)^2 = 3^2 = 9$$

$$20 : (20 - 15)^2 = 5^2 = 25$$

$$\text{Step 3: Sum} = 25 + 9 + 0 + 9 + 25 = 68$$

$$\text{Step 4: Population variance } \sigma^2 = \frac{68}{5} = 13.6$$

$$\text{Step 5: } \sigma = \sqrt{13.6} = 3.69$$

Simple explanation:

- Variance tells us the average squared distance from the mean.
- Standard deviation tells us the average distance from the mean (in original units).

Computational Formula (easier for large data):

$$\sigma^2 = \frac{\sum x^2}{n} - (\bar{x})^2$$

Using the same example:

$$\sum x = 75, \sum x^2 = 100 + 144 + 225 + 324 + 400 = 1193$$

$$\sigma^2 = \frac{1193}{5} - \left(\frac{75}{5}\right)^2 = 238.6 - 225 = 13.6$$

4.4.2 For Grouped Data

Formula:

$$\sigma^2 = \frac{\sum(f_i \times x_i^2)}{N} - \left(\frac{\sum(f_i \times x_i)}{N}\right)^2$$

Example:

Class	f_i	x_i	$f_i x_i$	$f_i x_i^2$
0-10	5	5	25	125
10-20	8	15	120	1800
20-30	12	25	300	7500
30-40	7	35	245	8575
40-50	3	45	135	6075
Total	35		825	24075

$$\sigma^2 = \frac{24075}{35} - \left(\frac{825}{35}\right)^2 = 687.86 - (23.57)^2 = 687.86 - 555.61 = 132.25$$

$$\sigma = \sqrt{132.25} = 11.50$$

4.5 Summary Table of Dispersion Measures

Measure	What it Tells Us	Good Points	Bad Points
Range	Total spread	Very simple	Affected by outliers
Quartile Deviation	Spread of middle 50%	Not affected by outliers	Ignores half the data
Mean Deviation	Average distance from center	Uses all data	Absolute values are tricky
Standard Deviation	Typical distance from mean	Most used, useful	Affected by outliers

5 Moments (Advanced Shape Descriptors)

Moments are like “descriptions” of a distribution. The first four moments tell us:

1. **First moment:** Where is the center? (Mean)
2. **Second moment:** How spread out? (Variance)
3. **Third moment:** Is it tilted? (Skewness)
4. **Fourth moment:** Is it peaked or flat? (Kurtosis)

5.1 Raw Moments (About the Origin)

The **r-th raw moment** is the average of the r-th power of values.

$$\mu'_r = \frac{x_1^r + x_2^r + \cdots + x_n^r}{n} \quad \text{or} \quad \mu'_r = \frac{\sum(f_i \cdot x_i^r)}{N}$$

Special Cases:

Moment	Formula	What it Tells Us
μ'_1	$\sum x/n$	The MEAN
μ'_2	$\sum x^2/n$	Average of squares
μ'_3	$\sum x^3/n$	Average of cubes
μ'_4	$\sum x^4/n$	Average of fourth powers

Example: Find raw moments for: 2, 4, 6, 8, 10

$$n = 5, \quad x_i : 2, 4, 6, 8, 10$$

$$\begin{aligned} \mu'_1 &= \frac{2 + 4 + 6 + 8 + 10}{5} = \frac{30}{5} = 6 \\ \mu'_2 &= \frac{4 + 16 + 36 + 64 + 100}{5} = \frac{220}{5} = 44 \\ \mu'_3 &= \frac{8 + 64 + 216 + 512 + 1000}{5} = \frac{1800}{5} = 360 \\ \mu'_4 &= \frac{16 + 256 + 1296 + 4096 + 10000}{5} = \frac{15664}{5} = 3132.8 \end{aligned}$$

5.2 Central Moments (About the Mean)

The **r-th central moment** is the average of the r-th power of deviations from the mean.

$$\mu_r = \frac{\sum (x_i - \bar{x})^r}{n} \quad \text{or} \quad \mu_r = \frac{\sum f_i (x_i - \bar{x})^r}{N}$$

Special Cases:

Moment	Formula	What it Tells Us
μ_1	$\sum (x_i - \bar{x})/n = 0$	Always zero!
μ_2	$\sum (x_i - \bar{x})^2/n$	VARIANCE (σ^2)
μ_3	$\sum (x_i - \bar{x})^3/n$	Used for SKEWNESS
μ_4	$\sum (x_i - \bar{x})^4/n$	Used for KURTOSIS

5.3 Converting Raw to Central Moments (Important!)

You don't always need to calculate deviations from mean. You can convert raw moments to central moments:

$$\begin{aligned}\mu_1 &= 0 \\ \mu_2 &= \mu'_2 - (\mu'_1)^2 \\ \mu_3 &= \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3 \\ \mu_4 &= \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4\end{aligned}$$

Example: For the data 2, 4, 6, 8, 10:

$$\mu'_1 = 6, \mu'_2 = 44, \mu'_3 = 360, \mu'_4 = 3132.8$$

$$\begin{aligned}\mu_1 &= 0 \\ \mu_2 &= 44 - 6^2 = 44 - 36 = 8 \\ \mu_3 &= 360 - 3 \times 44 \times 6 + 2 \times 6^3 = 360 - 792 + 432 = 0 \\ \mu_4 &= 3132.8 - 4 \times 360 \times 6 + 6 \times 44 \times 36 - 3 \times 6^4 \\ &= 3132.8 - 8640 + 9504 - 3888 = 108.8\end{aligned}$$

Verify μ_2 directly: Deviations from mean (6): -4, -2, 0, 2, 4

Squares: 16, 4, 0, 4, 16

Sum = 40, $\mu_2 = 40/5 = 8 \checkmark$

Interpretation:

- $\mu_2 = 8$ is the variance
- $\mu_3 = 0$ means the distribution is symmetric

6 Skewness (Is the Data Tilted?)

Skewness tells us if data is **tilted** to one side.

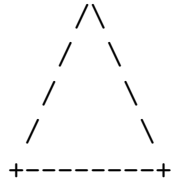
6.1 What is Skewness?

A distribution is:

- **Symmetric:** Left and right sides are mirror images
- **Positively skewed (right-skewed):** Tail is longer on the right
- **Negatively skewed (left-skewed):** Tail is longer on the left

6.1.1 Visual Comparison

Symmetric (Zero Skew):



Mean=Med=Mode

Positive Skew:



Mode<Med<Mean

Negative Skew:



Mean<Med<Mode

Real-life examples:

- **Positive skew:** Income distribution (most people earn moderate income, few earn very high)
- **Negative skew:** Age at retirement (most retire around a certain age, some retire early)

6.2 Measures of Skewness

6.2.1 Karl Pearson's Coefficient of Skewness

Formula:

$$Sk_p = \frac{\text{Mean} - \text{Mode}}{\text{Standard Deviation}}$$

Since $\text{Mode} \approx 3 \times \text{Median} - 2 \times \text{Mean}$, we can also use:

$$Sk_p = \frac{3(\text{Mean} - \text{Median})}{\text{Standard Deviation}}$$

Interpretation:

- $Sk_p = 0$: Symmetric
- $Sk_p > 0$: Positive skewness
- $Sk_p < 0$: Negative skewness
- Larger $|Sk_p|$ means more skewness

Example 1: Mean = 45, Median = 42, SD = 8

$$Sk_p = \frac{3(45 - 42)}{8} = \frac{9}{8} = 1.125$$

Interpretation: Positive skewness (right-skewed), moderate to high.

Example 2: Mean = 60, Mode = 65, SD = 10

$$Sk_p = \frac{60 - 65}{10} = -0.5$$

Interpretation: Negative skewness (left-skewed), mild.

6.2.2 Bowley's Coefficient of Skewness (Quartile-based)

This is used when you have quartiles and want to avoid extreme values.

Formula:

$$Sk_b = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

where Q_1 , Q_2 , Q_3 are the first, second (median), and third quartiles.

Interpretation:

- $Sk_b = 0$: Symmetric
- $Sk_b > 0$: Positive skewness
- $Sk_b < 0$: Negative skewness
- Value is always between -1 and $+1$

Example: $Q_1 = 25$, $Q_2 = 35$, $Q_3 = 50$

$$Sk_b = \frac{50 + 25 - 2 \times 35}{50 - 25} = \frac{75 - 70}{25} = \frac{5}{25} = 0.2$$

Interpretation: Slight positive skewness.

6.2.3 Moment-based Measure of Skewness

Using the third central moment:

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$
$$\gamma_1 = \frac{\mu_3}{\sigma^3} \quad (\text{coefficient of skewness})$$

Interpretation:

- $\gamma_1 = 0$: Symmetric
- $\gamma_1 > 0$: Positive skewness
- $\gamma_1 < 0$: Negative skewness

Example: $\mu_2 = 4$, $\mu_3 = 6$

$$\sigma = \sqrt{4} = 2, \quad \beta_1 = \frac{6^2}{4^3} = \frac{36}{64} = 0.5625, \quad \gamma_1 = \frac{6}{2^3} = \frac{6}{8} = 0.75$$

Interpretation: Positive skewness.

Example: $\mu_2 = 9$, $\mu_3 = -12$

$$\sigma = 3, \quad \beta_1 = \frac{(-12)^2}{9^3} = \frac{144}{729} = 0.1975, \quad \gamma_1 = \frac{-12}{27} = -0.444$$

Interpretation: Negative skewness.

6.3 Worked Example: Computing Skewness

Data: 4, 8, 12, 15, 20, 22, 28

Step 1: Find mean, median, mode, standard deviation

Sorted: 4, 8, 12, 15, 20, 22, 28

$$n = 7$$

$$\text{Mean} = \frac{4 + 8 + 12 + 15 + 20 + 22 + 28}{7} = \frac{109}{7} = 15.57$$

$$\text{Median} = 4\text{th value} = 15$$

$$\text{Mode} \approx 3 \times \text{Median} - 2 \times \text{Mean} = 45 - 31.14 = 13.86$$

Step 2: Find variance

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
4	-11.57	133.89
8	-7.57	57.32
12	-3.57	12.75
15	-0.57	0.33
20	4.43	19.62
22	6.43	41.33
28	12.43	154.48
Sum	0	419.72

$$\sigma^2 = \frac{419.72}{7} = 59.96, \quad \sigma = \sqrt{59.96} = 7.74$$

Step 3: Find μ_3

x_i	$(x_i - \bar{x})^3$
4	-1549.26
8	-433.97
12	-45.54
15	-0.19
20	86.88
22	265.72
28	1920.04
Sum	243.68

$$\mu_3 = \frac{243.68}{7} = 34.81$$

Step 4: Compute skewness measures

$$\text{Karl Pearson's: } Sk_p = \frac{15.57 - 13.86}{7.74} = 0.221$$

$$\text{Moment-based: } \gamma_1 = \frac{34.81}{(7.74)^3} = 0.075$$

Interpretation: Both measures show slight positive skewness (values close to zero).

7 Kurtosis (Is the Data Peaked or Flat?)

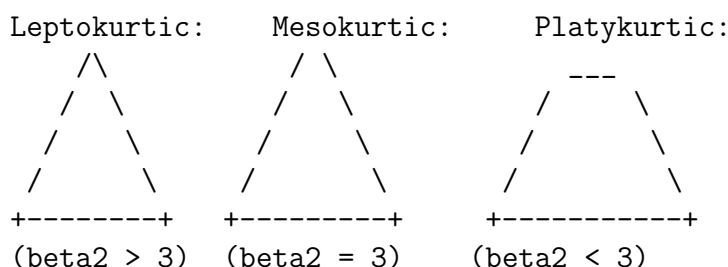
Kurtosis tells us if data is **peaked** or **flat** compared to a normal distribution.

7.1 What is Kurtosis?

Three types of kurtosis:

1. **Mesokurtic:** Like a normal distribution ($\beta_2 = 3$)
2. **Leptokurtic:** More peaked than normal ($\beta_2 > 3$)
3. **Platykurtic:** Flatter than normal ($\beta_2 < 3$)

7.1.1 Visual Comparison



Simple explanation:

- **Leptokurtic:** Sharp peak, heavy tails (more outliers)
- **Mesokurtic:** Normal peak
- **Platykurtic:** Flat peak, light tails (fewer outliers)

7.2 Measures of Kurtosis

7.2.1 Moment-based Measure

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{\mu_4}{\sigma^4} \quad \gamma_2 = \beta_2 - 3 \quad (\text{excess kurtosis})$$

Interpretation:

- $\beta_2 = 3$ (or $\gamma_2 = 0$): Mesokurtic
- $\beta_2 > 3$ (or $\gamma_2 > 0$): Leptokurtic
- $\beta_2 < 3$ (or $\gamma_2 < 0$): Platykurtic

Example 1: $\mu_2 = 16$, $\mu_4 = 400$

$$\beta_2 = \frac{400}{16^2} = \frac{400}{256} = 1.5625, \quad \gamma_2 = 1.5625 - 3 = -1.4375$$

Interpretation: Platykurtic ($\beta_2 < 3$).

Example 2: $\mu_2 = 25$, $\mu_4 = 2500$

$$\beta_2 = \frac{2500}{25^2} = \frac{2500}{625} = 4$$

Interpretation: Leptokurtic ($\beta_2 > 3$).

x	x^2	x^3	x^4
2	4	8	16
5	25	125	625
7	49	343	2401
9	81	729	6561
11	121	1331	14641
13	169	2197	28561
15	225	3375	50625
17	289	4913	83521
19	361	6859	130321
98	1324	19880	317272

7.3 Worked Example: Computing Both Skewness and Kurtosis

Data: 2, 5, 7, 9, 11, 13, 15, 17, 19

Step 1: Compute raw moments

$$n = 9$$

$$\mu'_1 = \frac{98}{9} = 10.89$$

$$\mu'_2 = \frac{1324}{9} = 147.11$$

$$\mu'_3 = \frac{19880}{9} = 2208.89$$

$$\mu'_4 = \frac{317272}{9} = 35252.44$$

Step 2: Compute central moments

$$\mu_2 = 147.11 - (10.89)^2 = 147.11 - 118.57 = 28.54$$

$$\begin{aligned}\mu_3 &= 2208.89 - 3 \times 147.11 \times 10.89 + 2 \times (10.89)^3 \\ &= 2208.89 - 4804.04 + 2582.15 = -13.00\end{aligned}$$

$$\begin{aligned}\mu_4 &= 35252.44 - 4 \times 2208.89 \times 10.89 + 6 \times 147.11 \times (10.89)^2 - 3 \times (10.89)^4 \\ &= 35252.44 - 96203.63 + 104641.37 - 42200.78 = 1489.40\end{aligned}$$

Step 3: Compute skewness

$$\sigma = \sqrt{28.54} = 5.34$$

$$\beta_1 = \frac{(-13)^2}{(28.54)^3} = \frac{169}{23237.7} = 0.0073 \quad \gamma_1 = \frac{-13}{(5.34)^3} = \frac{-13}{152.44} = -0.085$$

Step 4: Compute kurtosis

$$\beta_2 = \frac{1489.40}{(28.54)^2} = \frac{1489.40}{814.39} = 1.83 \quad \gamma_2 = 1.83 - 3 = -1.17$$

Interpretation:

- $\gamma_1 = -0.085$: Very slight negative skewness (nearly symmetric)
- $\beta_2 = 1.83$: $\beta_2 < 3$, so Platykurtic (flatter than normal)
- $\gamma_2 = -1.17$: Confirms platykurtic nature

8 Bivariate Data (Working with Two Variables)

So far, we've studied one variable at a time (univariate data). Now we look at **two variables** together (bivariate data).

8.1 Introduction

Bivariate data involves pairs of measurements for each observation.

Examples:

- Height and weight of students
- Hours studied and exam score
- Temperature and ice cream sales

Goals of bivariate analysis:

1. See if two variables are related
2. Measure the strength of the relationship
3. Predict one variable from the other

8.2 Scatter Diagram (Scatter Plot)

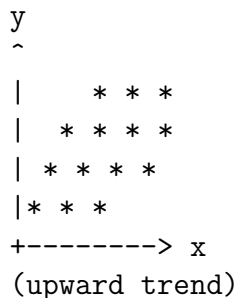
A scatter diagram is a simple graph showing the relationship between two variables.

How to make one:

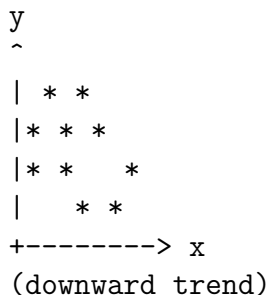
1. Put x-variable on horizontal axis
2. Put y-variable on vertical axis
3. Plot each pair (x, y) as a point

8.2.1 Interpreting Scatter Diagrams

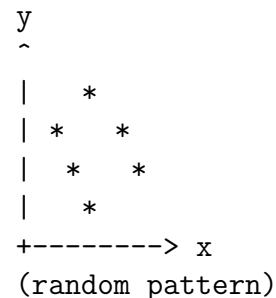
Positive Correlation:



Negative Correlation:



No Correlation:



Types of relationships:

- **Positive:** As x increases, y increases
- **Negative:** As x increases, y decreases
- **Linear:** Points follow a straight line
- **Non-linear:** Points follow a curve
- **No relationship:** No pattern

8.3 Bivariate Frequency Distribution

When there's a lot of data, we organize it in a **bivariate frequency table**.

$x \setminus y$	y_1	y_2	$\dots$	y_m	Row Total
x_1	f_{11}	f_{12}	$\dots$	f_{1m}	$f_{1\cdot}$
x_2	f_{21}	f_{22}	$\dots$	f_{2m}	$f_{2\cdot}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
x_n	f_{n1}	f_{n2}	$\dots$	f_{nm}	$f_{n\cdot}$
Col Total	$f_{\cdot 1}$	$f_{\cdot 2}$	$\dots$	$f_{\cdot m}$	N

Where f_{ij} = frequency in row i , column j ; $f_{i\cdot}$ = row total; $f_{\cdot j}$ = column total; N = grand total.

8.3.1 Structure of a Bivariate Frequency Table

8.4 Worked Example: Creating a Bivariate Table

Data: Marks of 30 students in Mathematics (x) and Statistics (y)

Step 1: Determine class intervals

Mathematics (x) :Min = 12, Max = 48 $\Rightarrow$ 10 – 20, 20 – 30, 30 – 40, 40 – 50

Statistics (y) :Min = 15, Max = 45 $\Rightarrow$ 10 – 20, 20 – 30, 30 – 40, 40 – 50

$x \setminus y$	10–20	20–30	30–40	40–50	Row Total
10–20					
20–30					
30–40					
40–50					
Col Total					30

Step 2: Create empty table

Step 3: Fill the table by tallying After processing all 30 students:

$x \setminus y$	10–20	20–30	30–40	40–50	Row Total
10–20	4	1	0	0	5
20–30	0	7	1	0	8
30–40	0	1	9	0	10
40–50	0	0	2	5	7
Col Total	4	9	12	5	30

8.5 Marginal Distributions

A **marginal distribution** shows the distribution of one variable ignoring the other.

x Class	Frequency	Relative Frequency
10–20	5	$5/30 = 0.167$
20–30	8	$8/30 = 0.267$
30–40	10	$10/30 = 0.333$
40–50	7	$7/30 = 0.233$
Total	30	1.000

Marginal distribution of x (from row totals):

y Class	Frequency	Relative Frequency
10–20	4	$4/30 = 0.133$
20–30	9	$9/30 = 0.300$
30–40	12	$12/30 = 0.400$
40–50	5	$5/30 = 0.167$
Total	30	1.000

Marginal distribution of y (from column totals):

8.6 Conditional Distributions

A **conditional distribution** shows the distribution of one variable for a specific class of the other variable.

Example: Distribution of y given x is in class 20–30.

From the row $x = 20\text{--}30$: y frequencies: 0, 7, 1, 0, row total = 8.

y Class	Frequency	Conditional Probability
10–20	0	$0/8 = 0.000$
20–30	7	$7/8 = 0.875$
30–40	1	$1/8 = 0.125$
40–50	0	$0/8 = 0.000$
Total	8	1.000

Interpretation: If a student scores 20–30 in Mathematics, there's an 87.5% chance they score 20–30 in Statistics.

8.7 Mean of Marginal Distributions

Using class marks (midpoints):

Mean of x :

$$\text{Mean of } x = \frac{940}{30} = 31.33$$

x Class	Midpoint (x)	Frequency (f)	$f \cdot x$
10–20	15	5	75
20–30	25	8	200
30–40	35	10	350
40–50	45	7	315
Total		30	940

y Class	Midpoint (y)	Frequency (f)	$f \cdot y$
10–20	15	4	60
20–30	25	9	225
30–40	35	12	420
40–50	45	5	225
Total		30	930

Mean of y :

$$\text{Mean of } y = \frac{930}{30} = 31.00$$

9 Correlation (How Two Variables Move Together)

Correlation measures how strongly two variables move together.

9.1 Definition

Correlation describes the degree and direction of relationship between two variables.

Key points:

- Correlation measures association, NOT causation
- Value ranges from -1 to $+1$
- Shows strength and direction of the linear relationship

9.2 Types of Correlation

9.2.1 Based on Direction

1. **Positive Correlation** ($r > 0$): Both variables move in same direction. As x increases, y increases. Example: Height and weight.
2. **Negative Correlation** ($r < 0$): Variables move in opposite directions. As x increases, y decreases. Example: Price and demand.
3. **Zero Correlation** ($r = 0$): No linear relationship. Example: Shoe size and IQ.

9.2.2 Based on Form

1. **Linear Correlation:** Points follow a straight line
2. **Non-linear Correlation:** Points follow a curve

9.3 Karl Pearson's Coefficient of Correlation

This is the most common measure of correlation.

Formula (computational):

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

9.3.1 Step-by-step Example 1

x	2	4	6	8	10
y	3	5	7	9	11

$$n = 5$$

$$\sum x = 2 + 4 + 6 + 8 + 10 = 30$$

$$\sum y = 3 + 5 + 7 + 9 + 11 = 35$$

$$\sum xy = 2 \cdot 3 + 4 \cdot 5 + 6 \cdot 7 + 8 \cdot 9 + 10 \cdot 11 = 6 + 20 + 42 + 72 + 110 = 250$$

$$\sum x^2 = 4 + 16 + 36 + 64 + 100 = 220$$

$$\sum y^2 = 9 + 25 + 49 + 81 + 121 = 285$$

$$\text{Numerator} = 5 \times 250 - 30 \times 35 = 1250 - 1050 = 200$$

$$\begin{aligned} \text{Denominator} &= \sqrt{(5 \times 220 - 30^2)(5 \times 285 - 35^2)} \\ &= \sqrt{(1100 - 900)(1425 - 1225)} = \sqrt{200 \times 200} = \sqrt{40000} = 200 \end{aligned}$$

$$r = \frac{200}{200} = 1.0$$

Interpretation: Perfect positive correlation ($r = +1$). The data lies exactly on a straight line.

9.3.2 Another Example

x	10	14	18	22	26	30
y	12	18	20	28	30	36

$$n = 6$$

$$\begin{aligned}\sum x &= 120, & \sum y &= 144 \\ \sum xy &= 3208, & \sum x^2 &= 2680, & \sum y^2 &= 3848\end{aligned}$$

$$\text{Numerator} = 6 \times 3208 - 120 \times 144 = 19248 - 17280 = 1968$$

$$\begin{aligned}\text{Denominator} &= \sqrt{(6 \times 2680 - 120^2)(6 \times 3848 - 144^2)} \\ &= \sqrt{(16080 - 14400)(23088 - 20736)} = \sqrt{1680 \times 2352} = \sqrt{3951360} = 1987.80\end{aligned}$$

$$r = \frac{1968}{1987.80} = 0.990$$

Interpretation: Very strong positive correlation ($r = 0.99$).

9.4 Correlation for Grouped Data (Bivariate Table)

Formula:

$$r = \frac{N \sum \sum f_{ij} x_i y_j - (\sum f_{i.} x_i)(\sum f_{.j} y_j)}{\sqrt{[N \sum f_{i.} x_i^2 - (\sum f_{i.} x_i)^2][N \sum f_{.j} y_j^2 - (\sum f_{.j} y_j)^2]}}$$

Using the bivariate table from earlier (reproduced below):

$x \setminus y$	10-20	20-30	30-40	40-50	$f_{i.}$
10-20	4	1	0	0	5
20-30	0	7	1	0	8
30-40	0	1	9	0	10
40-50	0	0	2	5	7
$f_{.j}$	4	9	12	5	$N = 30$

Class marks: x_i : 15, 25, 35, 45 y_j : 15, 25, 35, 45.

$$\sum f_{i.} x_i = 5 \times 15 + 8 \times 25 + 10 \times 35 + 7 \times 45 = 940$$

$$\sum f_{i.} x_i^2 = 5 \times 225 + 8 \times 625 + 10 \times 1225 + 7 \times 2025 = 32550$$

$$\sum f_{.j} y_j = 4 \times 15 + 9 \times 25 + 12 \times 35 + 5 \times 45 = 930$$

$$\sum f_{.j} y_j^2 = 4 \times 225 + 9 \times 625 + 12 \times 1225 + 5 \times 2025 = 31350$$

$$\sum \sum f_{ij} x_i y_j = 31700 \quad (\text{computed from table})$$

$$\begin{aligned}
\text{Numerator} &= 30 \times 31700 - 940 \times 930 = 951000 - 874200 = 76800 \\
\text{Denominator} &= \sqrt{(30 \times 32550 - 940^2)(30 \times 31350 - 930^2)} \\
&= \sqrt{(976500 - 883600)(940500 - 864900)} \\
&= \sqrt{92900 \times 75600} = 83804.8 \\
r &= \frac{76800}{83804.8} = 0.916
\end{aligned}$$

Interpretation: Strong positive correlation ($r = 0.916$) between Mathematics and Statistics marks.

9.5 Properties of Correlation Coefficient

1. **Range:** $-1 \leq r \leq +1$
2. **Unit-free:** r doesn't depend on units
3. **Symmetric:** $r(x, y) = r(y, x)$
4. **Linear measure:** Only measures linear relationships
5. $r = +1$: Perfect positive correlation
6. $r = -1$: Perfect negative correlation
7. $r = 0$: No linear correlation

9.6 Interpreting r Values

Value of r	Interpretation
$r = +1$	Perfect positive correlation
+0.8 to +0.99	Very strong positive correlation
+0.6 to +0.79	Strong positive correlation
+0.4 to +0.59	Moderate positive correlation
+0.2 to +0.39	Weak positive correlation
0 to +0.19	Very weak or no correlation
0	No linear correlation
-0.01 to -0.19	Very weak negative correlation
-0.2 to -0.39	Weak negative correlation
-0.4 to -0.59	Moderate negative correlation
-0.6 to -0.79	Strong negative correlation
-0.8 to -0.99	Very strong negative correlation
$r = -1$	Perfect negative correlation

9.7 Important Caveats

1. **Correlation does NOT imply causation.**
 - Example: Ice cream sales and drowning are correlated (both increase in summer), but ice cream doesn't cause drowning!

2. **Outliers can affect r significantly.**
3. **r measures only linear relationships** – two variables could have a perfect curve relationship and r could be zero.
4. **Spurious correlation:** Two variables may appear related by chance.

9.8 Coefficient of Determination

r^2 is called the **coefficient of determination**.

- r^2 represents the proportion of variance in y explained by x .
- Range: $0 \leq r^2 \leq 1$.
- If $r = 0.8$, $r^2 = 0.64 \rightarrow 64\%$ of variation in y is explained by x .

10 Rank Correlation (When Data Has Ranks)

When data is in the form of ranks (ordinal data), we use Spearman's rank correlation.

10.1 When to Use Rank Correlation

- Data is ordinal (rankings, preferences)
- Data doesn't follow normal distribution
- Sample size is small
- Only relative order is known, not actual values

10.2 Spearman's Rank Correlation Coefficient

Formula (no tied ranks):

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

where d = difference between ranks of the same observation, n = number of paired observations.

10.2.1 Step-by-step Example

Data: Two judges ranked 8 participants:

Participant	Judge A Rank	Judge B Rank
1	3	4
2	1	2
3	6	5
4	8	7
5	2	1
6	4	6
7	5	3
8	7	8

Step 1: Find differences d and d^2

Participant	A Rank	B Rank	$d = A - B$	d^2
1	3	4	-1	1
2	1	2	-1	1
3	6	5	1	1
4	8	7	1	1
5	2	1	1	1
6	4	6	-2	4
7	5	3	2	4
8	7	8	-1	1
			$\sum d^2$	14

Step 2: Apply formula

$$r_s = 1 - \frac{6 \times 14}{8(8^2 - 1)} = 1 - \frac{84}{8 \times 63} = 1 - \frac{84}{504} = 1 - 0.1667 = 0.8333$$

Interpretation: Strong positive rank correlation ($r_s = 0.833$). The two judges generally agree.

10.3 Rank Correlation with Tied Ranks

When values are equal, we assign **average ranks**.

Example of average rank assignment: Values: 10, 12, 12, 15, 18

Ranks: 10 $\rightarrow$ 1; two 12s tied for positions 2 and 3 $\rightarrow$ each gets $(2 + 3)/2 = 2.5$; 15 $\rightarrow$ 4; 18 $\rightarrow$ 5.

Example with ties: Two teachers graded 8 scripts:

Script	Teacher A	Teacher B
1	85	80
2	75	75
3	85	78
4	70	72
5	90	85
6	75	70
7	80	82
8	75	76

Step 1: Assign ranks with ties Teacher A scores: 70, 75, 75, 75, 80, 85, 85, 90

70 $\rightarrow$ 1

75 (three values, positions 2, 3, 4) $\rightarrow$ each gets 3

80 $\rightarrow$ 5

85 (two values, positions 6, 7) $\rightarrow$ each gets 6.5

90 $\rightarrow$ 8

Teacher B scores: 70, 72, 75, 76, 78, 80, 82, 85 (no ties, ranks 1 to 8 in order).

Script	Rank A	Rank B	d	d^2
1	6.5	6	0.5	0.25
2	3	3	0	0
3	6.5	5	1.5	2.25
4	1	2	-1	1
5	8	8	0	0
6	3	1	2	4
7	5	7	-2	4
8	3	4	-1	1
			$\sum d^2$	12.5

Step 2: Find d and d^2

$$r_s = 1 - \frac{6 \times 12.5}{8 \times 63} = 1 - \frac{75}{504} = 0.8512$$

Interpretation: Strong positive rank correlation ($r_s = 0.851$).

10.4 When to Use Pearson vs Spearman

Situation	Pearson	Spearman
Data is normally distributed	Yes	Not necessary
Relationship is linear	Yes	Not necessary
Data is ordinal	No	Yes
Data has outliers	Sensitive	Robust
Only ranks available	Cannot use	Yes

11 Regression (Predicting One Variable from Another)

While correlation measures the **strength** of a relationship, regression models the **form** of the relationship to make predictions.

11.1 Introduction

Regression allows us to:

- Predict the value of one variable given the other
- Understand how much y changes when x changes
- Find the line (or curve) of best fit

Key terms:

- **Dependent variable (y):** What we want to predict
- **Independent variable (x):** What we use for prediction

11.2 Regression Lines

There are two regression lines:

1. **Regression line of y on x :** $y = a + bx$ (predict y from x)
2. **Regression line of x on y :** $x = a' + b'y$ (predict x from y)

These are different unless correlation is perfect ($r = \pm 1$).

11.3 Method of Least Squares

The method of least squares finds the line that minimizes the sum of squared vertical distances from the data points to the line.

11.3.1 Normal Equations

For y on x :

$$\sum y = n a + b \sum x \quad (1)$$

$$\sum xy = a \sum x + b \sum x^2 \quad (2)$$

Solving for b and a :

$$b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}, \quad a = \bar{y} - b\bar{x}$$

For x on y :

$$b' = \frac{n \sum xy - \sum x \sum y}{n \sum y^2 - (\sum y)^2}, \quad a' = \bar{x} - b'\bar{y}$$

11.3.2 Step-by-step Example

Data:

x	2	4	6	8	10	12
y	5	7	10	12	14	18

$$n = 6$$

$$\begin{aligned} \sum x &= 42, & \sum y &= 66 \\ \sum xy &= 550, & \sum x^2 &= 364, & \sum y^2 &= 838 \\ \bar{x} &= 42/6 = 7, & \bar{y} &= 66/6 = 11 \end{aligned}$$

Regression of y on x :

$$\begin{aligned} b &= \frac{6 \times 550 - 42 \times 66}{6 \times 364 - 42^2} = \frac{3300 - 2772}{2184 - 1764} = \frac{528}{420} = 1.2571 \\ a &= 11 - 1.2571 \times 7 = 11 - 8.8 = 2.2 \\ y &= 2.2 + 1.2571 x \end{aligned}$$

Regression of x on y :

$$b' = \frac{6 \times 550 - 42 \times 66}{6 \times 838 - 66^2} = \frac{528}{5028 - 4356} = \frac{528}{672} = 0.7857$$

$$a' = 7 - 0.7857 \times 11 = 7 - 8.6429 = -1.6429$$

$$x = -1.6429 + 0.7857y$$

Relationship with correlation:

$$r = \sqrt{b \times b'} = \sqrt{1.2571 \times 0.7857} = \sqrt{0.9878} = 0.9939$$

Predictions:

- Estimate y when $x = 15$: $y = 2.2 + 1.2571 \times 15 = 2.2 + 18.8565 = 21.06$
- Estimate x when $y = 25$: $x = -1.6429 + 0.7857 \times 25 = -1.6429 + 19.6425 = 18.0$

11.4 Properties of Regression Coefficients

1. **Relationship with correlation:** $r = \sqrt{b_{yx} \cdot b_{xy}}$
2. **Sign consistency:** Both coefficients have the same sign as r .
3. **Unit-dependent:** Regression coefficients depend on units (unlike r).
4. **Intersection point:** Both lines pass through $(\bar{x}, \bar{y})$.

11.5 Correlation vs Regression Summary

Aspect	Correlation	Regression
Purpose	Measures strength and direction	Models relationship for prediction
Variables	Symmetric	One dependent, one independent
Coefficient	r (between -1 and $+1$)	b (no fixed range)
Unit-free?	Yes	No
Prediction	Not used directly	Used for prediction

12 Multiple Regression (Predicting with Many Variables)

In real life, the dependent variable often depends on **multiple** factors.

12.1 Introduction

Multiple linear regression extends simple regression to include two or more predictors.

General form:

$$y = a + b_1x_1 + b_2x_2 + \cdots + b_kx_k$$

For two independent variables:

$$y = a + b_1x_1 + b_2x_2$$

This represents a **regression plane** in 3D space.

12.2 Normal Equations (for 2 predictors)

Using least squares, we get:

$$\sum y = na + b_1 \sum x_1 + b_2 \sum x_2 \quad (3)$$

$$\sum x_1 y = a \sum x_1 + b_1 \sum x_1^2 + b_2 \sum x_1 x_2 \quad (4)$$

$$\sum x_2 y = a \sum x_2 + b_1 \sum x_1 x_2 + b_2 \sum x_2^2 \quad (5)$$

12.2.1 Solving Using Corrected Sums

First, compute corrected sums:

$$\begin{aligned} S_{11} &= \sum x_1^2 - \frac{(\sum x_1)^2}{n} \\ S_{22} &= \sum x_2^2 - \frac{(\sum x_2)^2}{n} \\ S_{12} &= \sum x_1 x_2 - \frac{(\sum x_1)(\sum x_2)}{n} \\ S_{1y} &= \sum x_1 y - \frac{(\sum x_1)(\sum y)}{n} \\ S_{2y} &= \sum x_2 y - \frac{(\sum x_2)(\sum y)}{n} \end{aligned}$$

Then solve:

$$S_{11} b_1 + S_{12} b_2 = S_{1y} \quad (6)$$

$$S_{12} b_1 + S_{22} b_2 = S_{2y} \quad (7)$$

And:

$$a = \bar{y} - b_1 \bar{x}_1 - b_2 \bar{x}_2$$

12.3 Worked Example

Problem: Predict exam score (y) from hours studied (x_1) and practice tests (x_2).

Student	x_1 (Hours)	x_2 (Tests)	y (Score)
1	3	2	75
2	4	3	80
3	5	2	82
4	3	4	78
5	6	3	88
6	7	4	92
7	4	2	76
8	5	5	85
9	6	4	90
10	7	3	86

Step 1: Compute sums

	x_1	x_2	y	x_1^2	x_2^2	x_1x_2	x_1y	x_2y
1	3	2	75	9	4	6	225	150
2	4	3	80	16	9	12	320	240
3	5	2	82	25	4	10	410	164
4	3	4	78	9	16	12	234	312
5	6	3	88	36	9	18	528	264
6	7	4	92	49	16	28	644	368
7	4	2	76	16	4	8	304	152
8	5	5	85	25	25	25	425	425
9	6	4	90	36	16	24	540	360
10	7	3	86	49	9	21	602	258
Sum	50	32	832	270	112	164	4232	2693

$$n = 10$$

$$\bar{x}_1 = 50/10 = 5, \quad \bar{x}_2 = 32/10 = 3.2, \quad \bar{y} = 832/10 = 83.2$$

Step 2: Compute corrected sums

$$S_{11} = 270 - 50^2/10 = 270 - 250 = 20$$

$$S_{22} = 112 - 32^2/10 = 112 - 102.4 = 9.6$$

$$S_{12} = 164 - 50 \times 32/10 = 164 - 160 = 4$$

$$S_{1y} = 4232 - 50 \times 832/10 = 4232 - 4160 = 72$$

$$S_{2y} = 2693 - 32 \times 832/10 = 2693 - 2662.4 = 30.6$$

Step 3: Solve for b_1 and b_2

$$20b_1 + 4b_2 = 72$$

$$4b_1 + 9.6b_2 = 30.6$$

Using Cramer's rule:

$$\Delta = 20 \times 9.6 - 4^2 = 192 - 16 = 176$$

$$b_1 = \frac{72 \times 9.6 - 30.6 \times 4}{176} = \frac{691.2 - 122.4}{176} = \frac{568.8}{176} = 3.2318$$

$$b_2 = \frac{30.6 \times 20 - 72 \times 4}{176} = \frac{612 - 288}{176} = \frac{324}{176} = 1.8409$$

Step 4: Find intercept

$$a = 83.2 - 3.2318 \times 5 - 1.8409 \times 3.2 = 83.2 - 16.159 - 5.8909 = 61.1501$$

Step 5: Regression equation

$$\hat{y} = 61.15 + 3.23x_1 + 1.84x_2$$

Interpretation:

- $b_1 = 3.23$: Holding x_2 constant, each additional hour of study increases score by 3.23 points.
- $b_2 = 1.84$: Holding x_1 constant, each additional practice test increases score by 1.84 points.

Step 6: Make a prediction

$$\hat{y} = 61.15 + 3.23 \times 8 + 1.84 \times 5 = 61.15 + 25.84 + 9.20 = 96.19$$

12.4 Coefficient of Multiple Determination (R^2)

R^2 measures how well the model fits the data.

For our example:

$$\begin{aligned}SST &= \sum (y_i - \bar{y})^2 = 315.6 \\SSR &= b_1 S_{1y} + b_2 S_{2y} = 3.2318 \times 72 + 1.8409 \times 30.6 = 289.02 \\R^2 &= \frac{SSR}{SST} = \frac{289.02}{315.6} = 0.9158\end{aligned}$$

Interpretation: 91.6% of the variation in exam scores is explained by hours studied and number of practice tests.

12.5 Assumptions of Multiple Regression

1. **Linearity:** Relationship between y and each x is linear.
2. **Independence:** Observations are independent.
3. **Homoscedasticity:** Constant variance of errors.
4. **Normality:** Errors are normally distributed.
5. **No multicollinearity:** Independent variables aren't highly correlated.

Multicollinearity occurs when x_1 and x_2 are highly correlated. This makes coefficient estimates unstable.

13 Summary

13.1 Key Concepts

Concept	What It Tells Us	Key Measure
Central Tendency	Where is the center?	Mean, Median, Mode
Dispersion	How spread out?	Standard Deviation, Variance
Moments	Shape descriptors	$\mu_1, \mu_2, \mu_3, \mu_4$
Skewness	Is it tilted?	γ_1
Kurtosis	Is it peaked/flat?	γ_2
Correlation	How strongly related?	r
Regression	How to predict?	b
Multiple Regression	Predict with many vars	$b_1, b_2, \dots$

13.2 Quick Reference Formulas

This guide covers all the topics from your lecture notes in a beginner-friendly way. Take it step by step, and don't worry if some concepts take time to understand. Statistics builds on itself, so each concept connects to the others.

Remember:

Measure	Formula
Mean (raw)	$\bar{x} = \sum x/n$
Variance (raw)	$\sigma^2 = \sum (x - \bar{x})^2/n$
Standard Deviation	$\sigma = \sqrt{\sigma^2}$
Karl Pearson's Skewness	$Sk_p = 3(\bar{x} - \text{Median})/\sigma$
Moment Skewness	$\gamma_1 = \mu_3/\sigma^3$
Kurtosis	$\gamma_2 = \mu_4/\sigma^4 - 3$
Correlation	$r = \text{cov}(x, y)/(\sigma_x \sigma_y)$
Regression (y on x)	$y = a + bx, \quad b = r \sigma_y/\sigma_x$
Rank Correlation	$r_s = 1 - 6 \sum d^2/[n(n^2 - 1)]$

- Practice with examples
- Ask questions when stuck
- Connect concepts to real life
- One step at a time!

Good luck with your studies!